Off-centre CoE cosmology: the mathematical workflow from geometry to the Planck TT spectrum

Working notes for C. A. (compiled 23 October 2026)

Abstract

These notes set out, in logical order, the mathematics and physics used in the off-centre "CoE" cosmology workflow: (i) the geometry (off-centre observer, vision radius, no Λ) and its volume history; (ii) density laws and the Migkas $A/A_{\rm all}$ fit; (iii) spherical-harmonic primers and the inverse problem of building a target angular spectrum from clumps and voids; (iv) a realistic void/cluster web; (v) a stochastic density field projected onto the sky; (vi) the acoustic fit and the decomposition of its residuals; (vii) the damped half-scale model and its corrected projection; (viii) a parameter glossary and a statement of which quantities are predicted and which are fitted. Goodness-of-fit is reported as χ^2 (Planck errors unless stated) and Pearson r^2 between data and model. No pass/fail judgement is made here.

1 Geometry of the CoE universe

Set-up. The observer sits off-centre at distance D from the centre of expansion (CoE). Working values: $D \simeq 58.2\,\mathrm{Gpc}$, vision (horizon) radius R explored at 4, 8 and 14 Gpc, and an outer edge of the matter shell at $1.25D \simeq 72.83\,\mathrm{Gpc}$. There is no cosmological constant: the dynamics are matter (plus radiation at early times) in a Newtonian/LTB description.

Volumes. The three volumes tracked are the observer's sphere $V_R = \frac{4}{3}\pi R^3$, the sphere out to the observer plus horizon $V_{D+R} = \frac{4}{3}\pi(D+R)^3$, and the outer shell. Today $V_R \simeq 268\,\mathrm{Gpc}^3$ and $V_{D+R} \simeq 1.0\times10^6\,\mathrm{Gpc}^3$, so $V_R/V_{D+R} \simeq 0.027\%$ and $V_R/V_{\rm shell} \simeq 0.017\%$.

Time dependence (assumed). The growth law is *assumed*, not derived from the geometry. Lengths follow the scale factor $a(t)$ with radiation + matter,

$$\left(\frac{\dot a}{a}\right)^2 \propto \frac{a_{\rm eq}}{a^4} + \frac{1}{a^3}, \qquad a_{\rm eq} = \frac{1}{3400}, \tag{1}$$

normalised so that $t_0 = 13.8\,\mathrm{Gyr}$. Without Λ this places equality at $\approx 41\,\mathrm{kyr}$ and recombination at $\approx 297\,\mathrm{kyr}$. In the pure phases $a \propto t^{1/2}$ (radiation) and $a \propto t^{2/3}$ (matter); the horizon grows as $R \propto t$. For a volume $V \propto t^n$ the fractional rate is

$$\dot V/V = n/t, \tag{2}$$

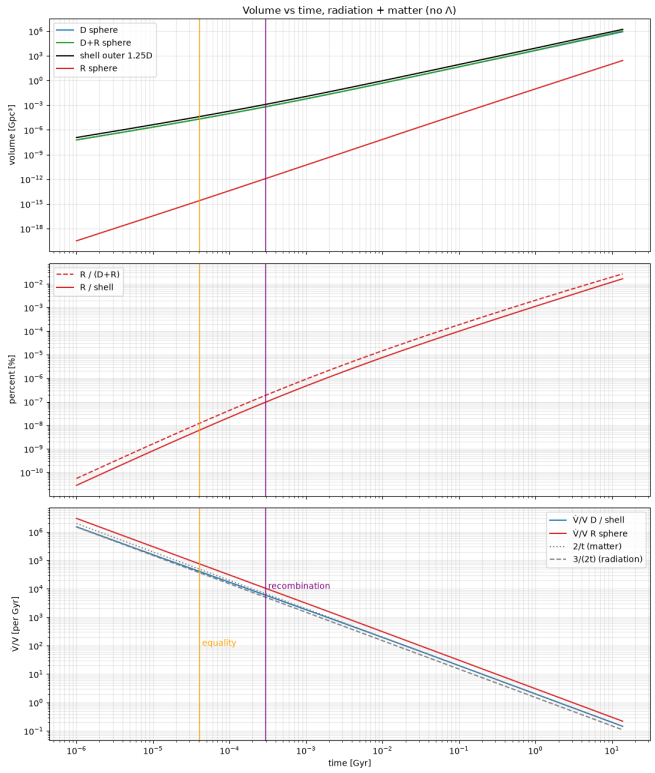


Figure 1: Volume versus time with radiation + matter (no Λ). Top: absolute volumes of the D sphere, $D{+}R$ sphere and $1.25D$ outer shell from $a(t)$ of Eq. (1). Middle: V_R as a percentage of the larger volumes. Bottom: fractional growth $\dot V/V = n/t$ for each volume. Vertical lines mark equality (≈41 kyr) and recombination (≈297 kyr).

so $n = 3/2$ (radiation era), $n = 2$ for D in the matter era, $n = 3$ for R. The fractional rate is therefore largest at early times and falls as $1/t$; for D today it is $\approx 0.145\,\mathrm{Gyr}^{-1}$. This is ordinary decelerating expansion put in by assumption (Fig. 1).

2 Density laws and the Migkas fit

Observable. Migkas et al. measure the cluster scaling-relation normalisation A in sky directions relative to the all-sky value, $A/A_{\rm all}$, binned in galactic longitude l (71 bins). In CoE the direction dependence arises because lines of sight at different angles ψ to the CoE axis traverse different matter densities and recession velocities.

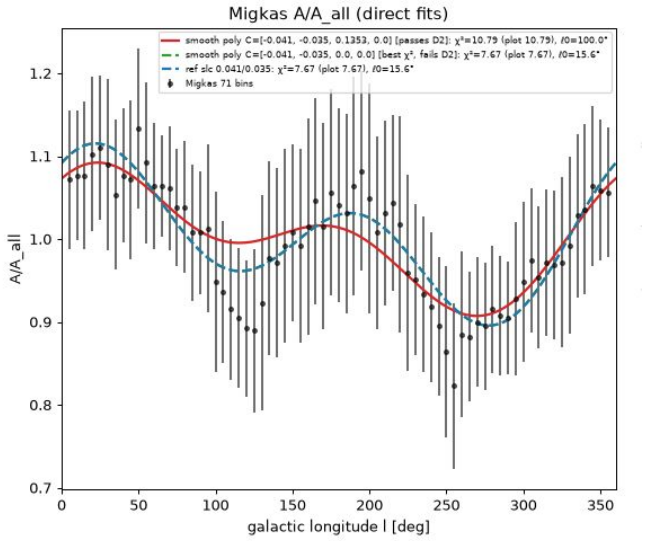


Figure 2: Migkas $A/A_{\rm all}$ (71 bins, black) against galactic longitude with CoE direct fits. Red: smooth polynomial in $\cos\psi$, $\chi^2 = 10.79$, $l_0 = 100°$. Green dashed: best-χ^2 variant, $\chi^2 = 7.67$, $l_0 = 15.6°$. Blue dashed: reference sinusoid (amplitudes 0.041/0.035), $\chi^2 = 7.67$. The two humps follow from projecting the axis direction $\psi(l)$ through Eq. (3).

Line-of-sight averaging. For a radial density law $\rho(r)$ about the CoE, a point at distance s along direction $\hat{n}$ from the observer has $r(s,\hat{n}) = |\mathbf{D} + s\hat{n}|$. The predicted anisotropy is the average over the cluster depth distribution $p(s)$,

$$\frac{A}{A_{\rm all}}(\hat{n}) = \frac{\int ds\, p(s)\, f[\rho(r(s,\hat{n}))]}{\langle \int ds\, p(s)\, f[\rho]\rangle_{\rm sky}}, \qquad (3)$$

where f maps the local density/velocity to the scaling-relation shift. Density laws tried include exponential, η-form and level-set families (power-law, cored, Gaussian, soft).

CatWISE lock. The CatWISE/Secrest matter dipole constrains the inward/outward density contrast seen by the observer; the density law was locked to a $\sim$3% contrast.

Fit. $\chi^2 = \sum_i (y_i - m_i)^2/\sigma_i^2$ over the 71 Migkas bins. Fig. 2 shows the two-hump structure obtained: $\chi^2 = 10.79$ ($l_0 = 100°$) and $\chi^2 = 7.67$ ($l_0 = 15.6°$), the latter equal to a reference sinusoid fit ($\chi^2 = 7.67$).

3 Harmonic basics and the inverse problem

Any sky pattern $\Delta T(\hat{n})$ is expanded in spherical harmonics,

$$\Delta T(\hat{n}) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat{n}), \quad C_\ell = \frac{1}{2\ell+1}\sum_m |a_{\ell m}|^2. \quad (4)$$

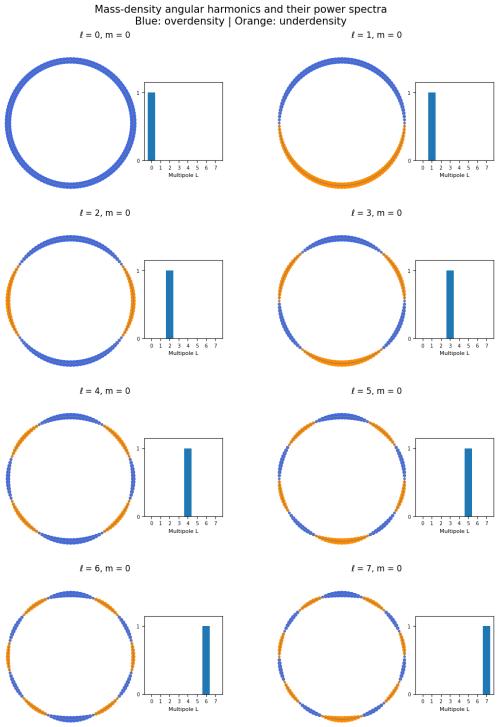


Figure 3: Single axisymmetric harmonics $\ell = 0\ldots7$, $m = 0$ drawn as density on a ring (blue over-, orange under-density) with the power bar at that ℓ: each pure harmonic puts all power in one multipole.

ℓ counts angular wavelengths, $\theta \approx 180°/\ell$. Axisymmetric ($m = 0$) patterns are Legendre polynomials $P_\ell(\cos\theta)$. For N equal point masses equally spaced on a ring, the angular Fourier power is non-zero only at multiples of N (aliasing). The primer figures (Figs. 3–6) build intuition for the *inverse problem*: given a target C_ℓ, which arrangement of over-densities (clumps, blue) and under-densities (voids, orange/red) produces it? Because C_ℓ discards phases, the inverse problem is degenerate: many mass arrangements give the same spectrum.

4 A realistic web of voids and clusters

To replace hand-placed masses, a realistic web is generated (Fig. 8): clusters above a mass cut and voids from a ΛCDM-like seed realisation, placed around the observer. Each structure contributes a potential Φ through Poisson's equation, and photons crossing the moving/evolving potentials pick up temperature shifts (integrated Sachs–Wolfe and moving-lens terms). The resulting D_ℓ is computed for $R = 4$, 8, 14 Gpc ("level-set ladders", Fig. 9), and Monte-Carlo webs and kernels are compared to Planck at low ℓ (Fig. 10). At the lowest multipoles, large "passing-universe" domains dominate (Fig. 10).

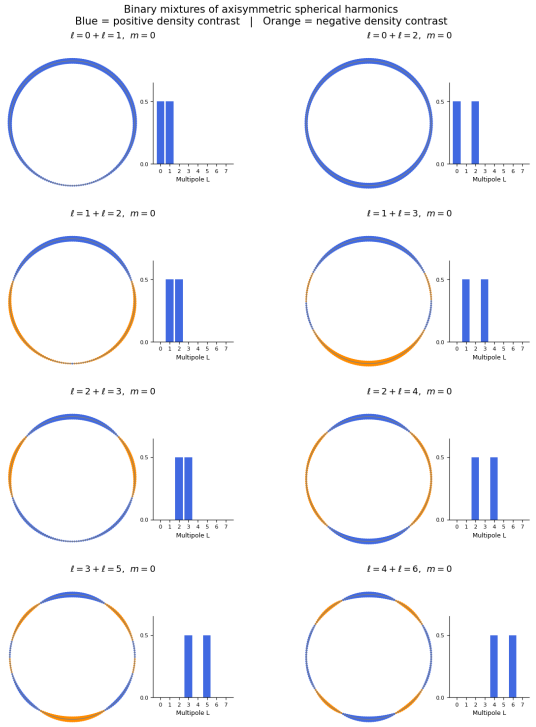


Figure 4: Binary mixtures $\ell_1 + \ell_2$ ($m = 0$): the density pattern is the sum of two Legendre modes and the spectrum has two bars. Sums of modes are how a structured target spectrum is assembled.

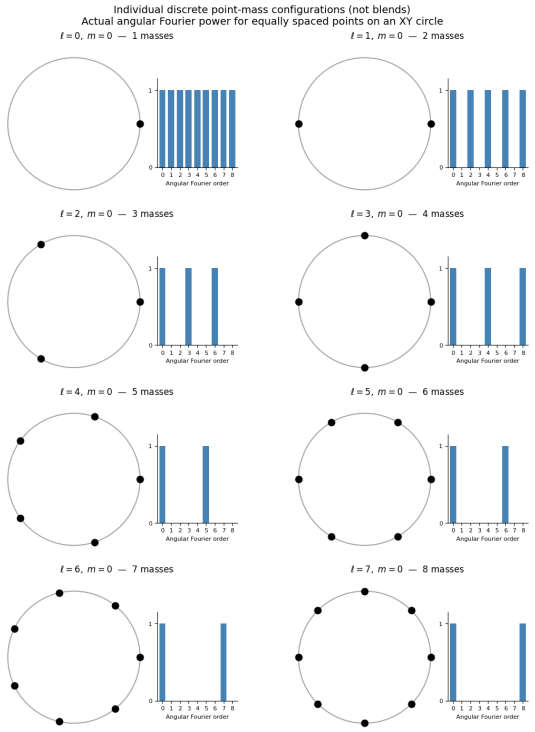


Figure 5: Discrete point-mass configurations: N equally spaced masses on a circle. The actual angular Fourier power appears at $N, 2N, 3N, \ldots$, showing that discrete masses alias power to harmonics of N rather than producing a single mode.

5 Stochastic field and sky projection

Source. Instead of discrete objects, the density contrast δ in a shell is treated as a Gaussian random field with power spectrum $P_\delta(k)$. The potential follows from Poisson,

$$\Phi_k = -\frac{4\pi G\rho\,\delta_k}{k^2} \;\Rightarrow\; P_\Phi = \frac{(4\pi G\rho)^2 P_\delta}{k^4}. \tag{5}$$

In the code the dominant temperature term is the moving-lens term $\Delta T/T \propto K_M(\chi)\,\mathbf{u}\cdot\nabla\Phi$; averaging $(\mathbf{k}\cdot\mathbf{u})^2$ over directions gives a factor $k^2/3$.

Projection. With radial window $W(\chi) = \exp[-(\chi - R_c)^2/2w^2]$ (shell centre R_c, width w), the exact projection is

$$C_\ell = \frac{2}{\pi}\int k^2 dk\, P(k)\left|\int d\chi\, W(\chi)K(\chi)j_\ell(k\chi)\right|^2, \tag{6}$$

with j_ℓ the spherical Bessel function. For a shell thin compared to its radius, $j_\ell(k\chi)$ peaks at $k\chi \approx \ell + \frac{1}{2}$ (Limber):

$$\begin{aligned} D_\ell = &\frac{\ell(\ell+1)T_0^2}{2\pi}\int d\chi \\ &\times \frac{[K_M W]^2}{\chi^2}\frac{k^2}{3}P_\Phi(k)\Big|_{k=\frac{\ell+1/2}{\chi}}, \end{aligned} \tag{7}$$

with $D_\ell \equiv \ell(\ell+1)C_\ell/2\pi$ in $\mu\mathrm{K}^2$. A length r on the shell maps to $\Delta\ell \approx 2\pi R_c/r$.

Cosmic variance and a single sky. Each C_ℓ estimated from one sky has $2\ell + 1$ modes, so $\Delta C_\ell/C_\ell = \sqrt{2/(2\ell+1)}$; this is the red/blue $\pm 1\sigma$ band. The jagged line is one realisation of $a_{\ell m}$ drawn from the model.

Feature size. The $\ell \approx 220$ hump corresponds to $\theta = 180°/220 = 0.818°$, i.e. a physical size $R\theta$ of 57.1, 114.2, 199.9 Mpc at $R = 4$, 8, 14 Gpc (52–63, 105–126, 183–220 Mpc for ℓ 240–200). For comparison, generic literature ranges: voids 10–100 Mpc (supervoids up to ∼300), superclusters ∼100–200 Mpc, BAO 147 Mpc.

6 Acoustic fit and residual decomposition

Acoustic source. To produce peaks, the source spectrum gets an oscillation at scale r_s:

$$P_\delta = P_0 k^n [1 + A\cos(kr_s + \varphi)]e^{-(k/k_D)^2}. \tag{8}$$

Through $\Delta\ell = 2\pi R_c/r_s$ this gives evenly spaced peaks. "free_W" lets R_c, w float. Result: $r_s = 81.9$ Mpc, $R_c = 3.944$ Gpc, $\chi^2_{2-2500} = 11{,}772$ (Planck errors), Pearson $r^2 = 0.871$; ΛCDM: 2,558, 0.983. Peaks: CoE 227, 530, 831, 1131, 1431 vs Planck 224, 535, 821, 1122, 1412 (Fig. 14).

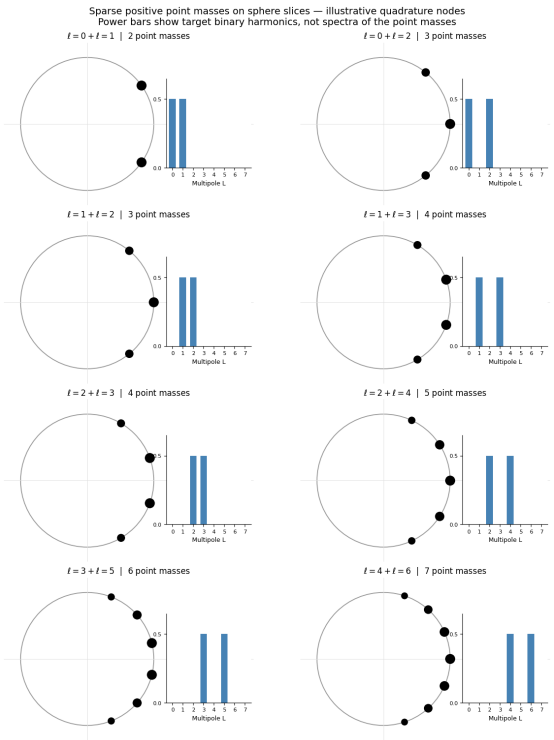


Figure 6: Sparse positive point masses on sphere slices used as quadrature nodes for target binary harmonics ($\ell = 0+\ell$). Bars show the target spectra, not the spectra of the point masses themselves; the figure illustrates sampling of a smooth mode with few masses.

Residuals. Define $\epsilon_\ell = (D_\ell^{\rm Planck} - D_\ell^{\rm CoE})/\sigma_\ell$. Two decompositions:

$$\text{undamped: } \epsilon_\ell = \textstyle\sum_j a_j \sin(2\pi j\ell/\Delta\ell + \phi_j), \tag{9}$$

$$\text{damped: } \epsilon_\ell = e^{-\ell/L} \textstyle\sum_j a_j \sin(2\pi j\ell/\Delta\ell + \phi_j). \tag{10}$$

Undamped plateaus at Pearson $r^2 = 0.551$. Damped with 1–9 harmonics: $r^2 = 0.797, 0.813, 0.831, 0.834, 0.835, 0.836, 0.836, 0.837, 0.837$; fundamental $\Delta\ell \approx 593$ (34.6σ), $L \approx 990$; starting amplitudes (σ) 34.6, 5.6, 5.7, 2.2, 1.3, 1.0, 1.2, 0.4, 0.5. Late troughs at $\ell \sim 1400$, 2000 are missed because the shared envelope has decayed.

Mapping to k. With $R \approx 3.94\,$Gpc: $r_2 = 2\pi R/593 \approx 42\,$Mpc $\approx r_s/2$, which modulates alternate peaks (odd/even alternation), and $k_B = L/R \approx 0.25\,\text{Mpc}^{-1}$ as the envelope scale (starting value).

7 Damped half-scale model

Model.

$$P_\delta = P_0 k^n \big[1 + A\cos(kr_s + \varphi) + Be^{-k/k_B}\cos(kr_2 + \varphi_2)\big] e^{-(k/k_D)^2}. \tag{11}$$

The B term is the minimal damped cosine (amplitude, decay, period, phase) suggested by the residuals. Free parameters (12): $P_0, n, A, r_s, \varphi, k_D, R_c, w, B, k_B, r_2, \varphi_2$; "half" variant ties $r_2 = r_s/2$ (11). All are floated in one χ^2 minimisation over $\ell = 2$–2500.

The $\ell = 50$ discontinuity. The code uses the exact projection (Eq. 6) for $\ell \le 50$ and Limber (Eq. 7) above, multiplied by a correction $c_\ell = C_\ell^{\rm exact}/C_\ell^{\rm Limber}$. That ratio had been computed for the earlier, narrower shell; when the fit widened the shell it was not recomputed, producing a 43% step at $\ell = 50$. Fix: compute the exact/Limber ratio at 69 nodes ($\ell = 50$–175 then every 40 to 2480), interpolate in $\log\ell$, apply at every $\ell > 50$, and recompute after each fit pass. The ℓ 50→51 change is now $\approx 1.1\%$, the curve's natural slope.

Results (r_2 free). $\log_{10} P_0 = -8.28$, $n = -0.183$, $A = 0.311$, $r_s = 79.5\,$Mpc, $\varphi = 1.37$, $k_D = 0.335\,\text{Mpc}^{-1}$, $R_c = 3.572\,$Gpc, $w = 0.336\,$Gpc, $B = 0.815$, $k_B = 0.138\,\text{Mpc}^{-1}$, $r_2 = 42.1\,$Mpc ($r_s/2 = 39.8$), $\varphi_2 = 3.64$.

Table 1: Fit statistics (Planck errors). Pearson r^2 for ℓ 2–2500.

Model	χ^2_{2-2500}	r^2	χ^2_{2-30}
CoE r_2 free	3,787	0.976	137
CoE half ($r_2 = r_s/2$)	4,131	0.966	49.1
CoE acoustic (no B)	11,772	0.871	–
ΛCDM	2,558	0.983	20.1

Table 2: Peaks: ℓ (D_ℓ, μK^2).

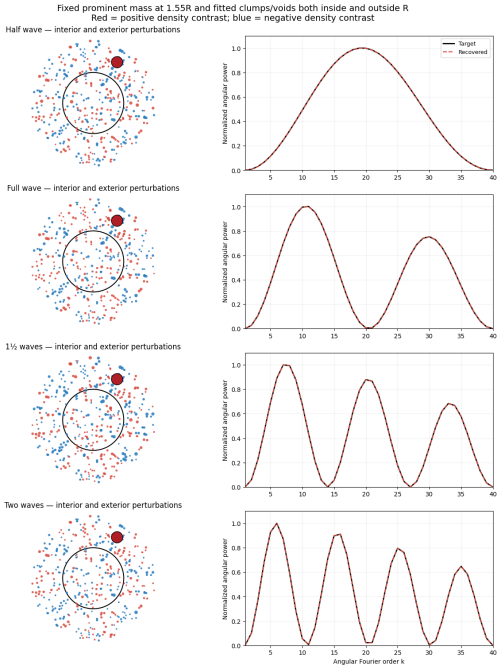
	Planck	CoE r_2 free	CoE half
1	228 (5726)	218 (5927)	207 (5223)
2	544 (2580)	531 (2449)	537 (2383)
3	823 (2525)	813 (2466)	818 (2433)
4	1121 (1239)	1130 (1293)	1132 (1260)
5	1417 (814)	1420 (822)	1426 (861)

The odd/even alternation is absent from the new residual. Remaining structure: a slow wave with $\Delta\ell \approx 1{,}384$ (± 8–11σ); ℓ = 2–3 raised to $\approx 3{,}300\,\mu$K^2 (off the top panel), which drives $\chi^2_{2-30} = 137$.

8 Parameters and status

Predicted vs fitted. From the geometry: the two-hump Migkas shape and ring placement given D, mass and density law (those three fitted). From the stochastic field: the spectrum shape follows from Eqs. (6)–(7) once the 12 source/shell parameters are fitted. Inverse-problem targets: the data require a second, damped mode at $r_2 \approx 42\,$Mpc $\approx r_s/2$ decaying at $k_B \approx 0.14\,\text{Mpc}^{-1}$; a physical shell model must supply this.

Next steps. (1) Careful numerics: exact projection throughout, low-ℓ control in the r_2-free fit, the $\Delta\ell \approx 1{,}384$ wave. (2) Analytic thin-shell projection: with W narrow,



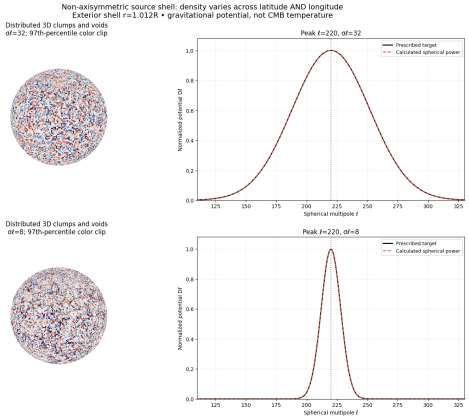


Figure 7: Inverse problem solved numerically. Left: a fixed prominent mass at $1.55R$ plus fitted clumps/voids inside and outside R reproduce target angular profiles (half wave, $1\frac{1}{2}$, $2\frac{1}{2}$ waves, two waves); red target vs fitted curve. Right: a non-axisymmetric source shell at $r = 1.012R$ (density varying in latitude and longitude; gravitational potential, not CMB temperature) built from distributed 3D clumps/voids reproduces a prescribed peak at $\ell = 220$ with widths $\sigma_\ell = 32$ and 8.

Table 3: Glossary.

D	observer–CoE distance (58.2 Gpc)
R	vision radius (4/8/14 Gpc)
$a_{\rm eq}$	scale factor at equality (1/3400)
P_0, n	source amplitude, tilt
A, r_s, φ	acoustic amplitude, scale, phase
k_D	small-scale (diffusion) cutoff
R_c, w	shell centre and Gaussian width
B, k_B	damped-mode amplitude, decay
r_2, φ_2	damped-mode scale, phase
$\Delta\ell, L$	residual period, envelope
χ^2	$\sum(D^{\rm P} - D^{\rm M})^2/\sigma^2$
Pearson r^2	squared correlation of $D^{\rm P}$, $D^{\rm M}$

$C_\ell \approx P(k_\ell)/R_c^2$ times a window transform, to see which of the 12 parameters the geometry fixes (e.g. whether r_s and $r_2 = r_s/2$ follow from the shell itself). (3) $D(t)$ from the Newtonian integrator instead of an assumed power law.

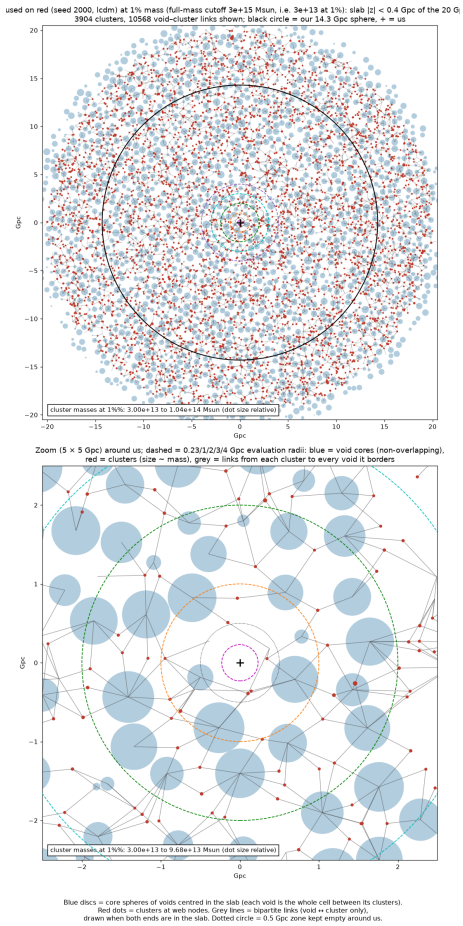


Figure 8: Realistic web: slab $|z| < 0.4\,\mathrm{Gpc}$ of a $20\,\mathrm{Gpc}$ box (seed 2000, 1% mass cut, 3904 clusters, 10568 void–cluster links); black circle is the $14.3\,\mathrm{Gpc}$ sphere, + the observer. Bottom: zoom ($\pm 9\,\mathrm{Gpc}$) showing voids (blue discs) and cluster links.

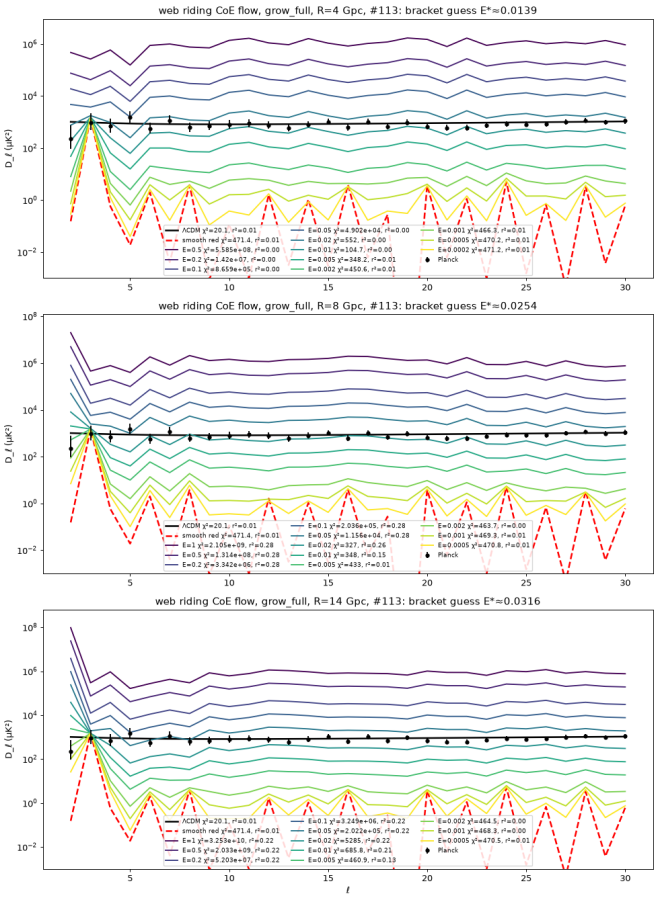
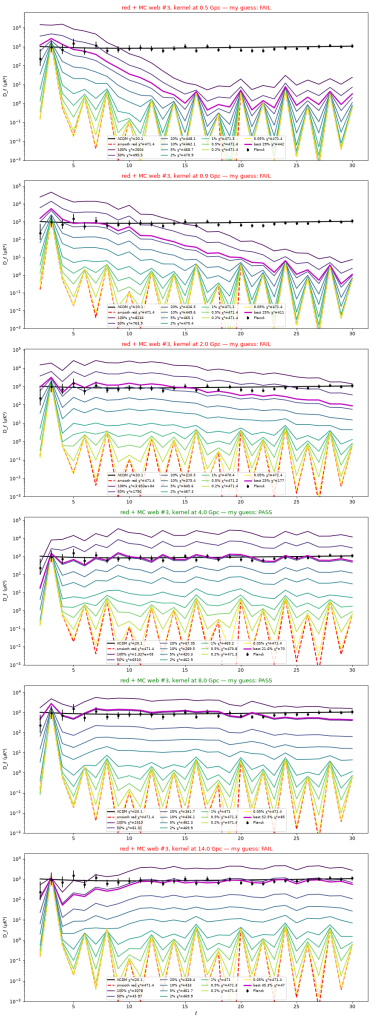


Figure 9: Web riding the CoE flow (grow_full, run #113) at $R = 4$, 8, $14\,\mathrm{Gpc}$: coloured curves are D_ℓ for a ladder of density level sets, black points Planck, red dashed the bracket guess E^*. The lines show how D_ℓ at $\ell \leq 30$ scales with the density threshold.



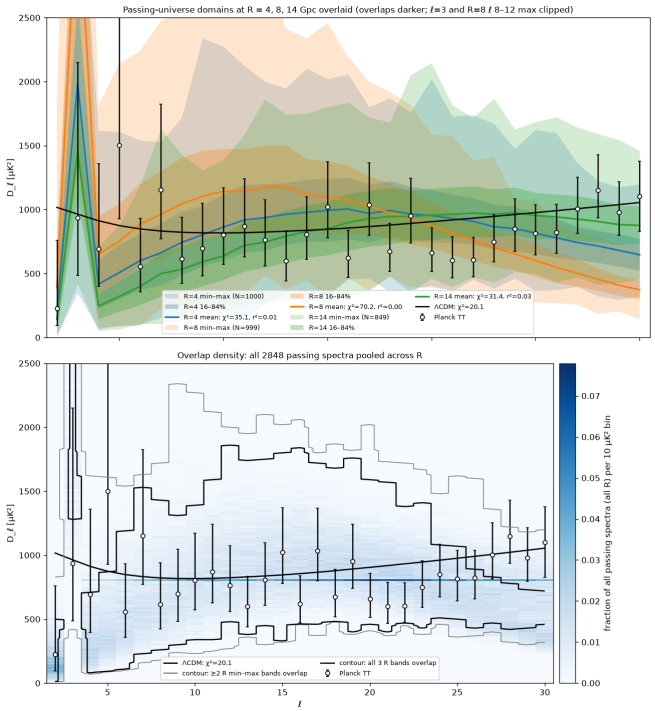


Figure 10: Left: red + Monte-Carlo webs with smoothing kernels 0.5–2 Gpc; each panel shows level-set ladders against Planck (black) with an in-panel visual guess. Right: passing-universe domains at $R = 4$, 8, 14 Gpc overlaid (top, bands are spreads over realisations; overlaps darker; $\ell = 3$ and $R = 8$ ℓ 8–12 maxima clipped) and the density of sky-plane domains (bottom) against Planck.

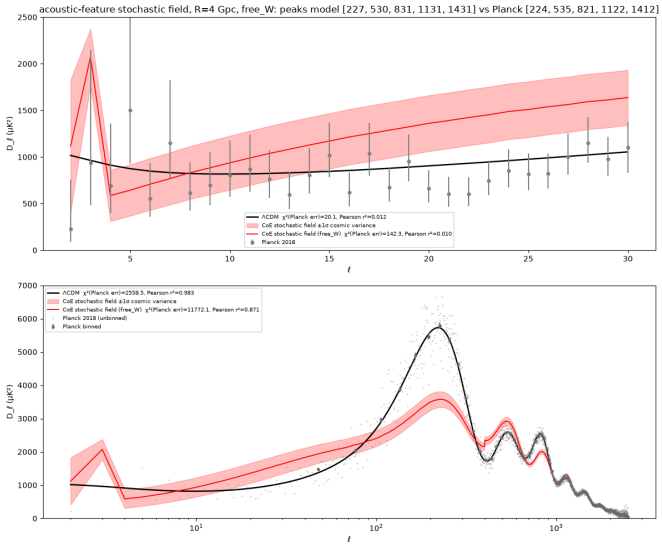


Figure 11: Stochastic field at $R = 4$ Gpc, $\lambda = 100$ ($\ell \leq 30$ fit with background, "l30_bg"). Blue band: $\pm 1\sigma$ cosmic variance of the field; jagged line: one-sy realisation; black: ΛCDM; points: Planck. Upper panel low ℓ, lower panel to high ℓ.

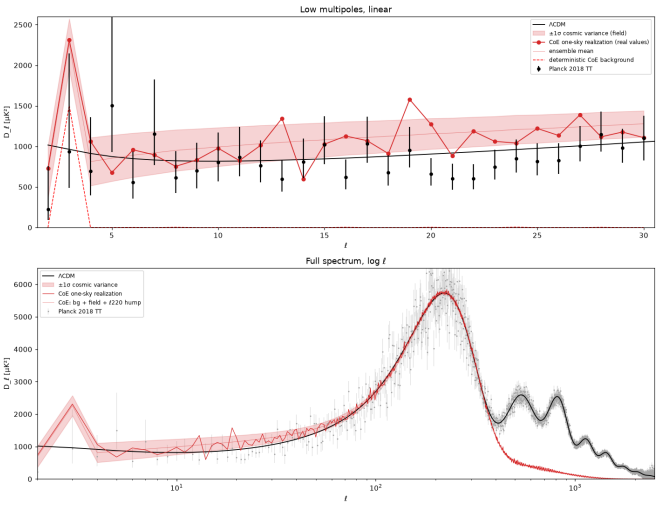


Figure 12: Low multipoles (linear axes): ΛCDM (black), Planck (grey), CoE one-sky realisation (red, real values) in the $\pm 1\sigma$ cosmic-variance band with faint ensemble mean. Bottom: full spectrum.

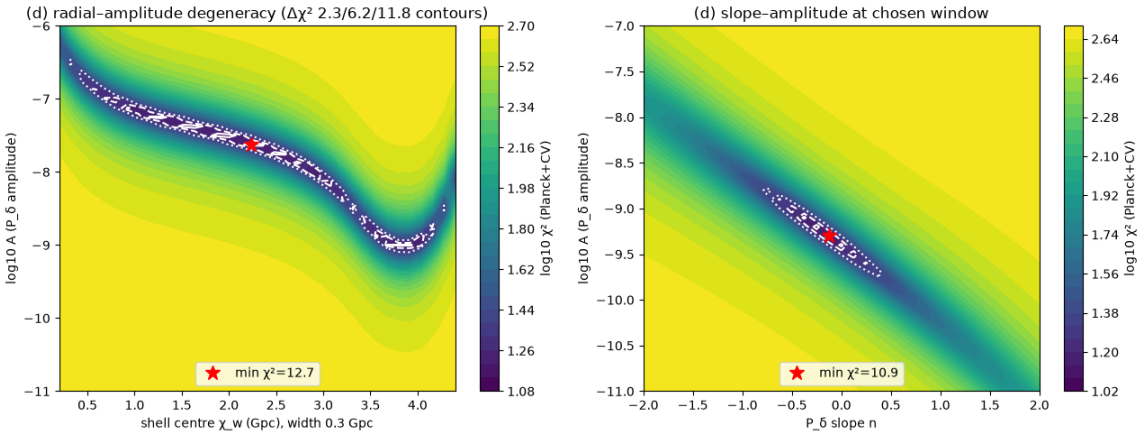


Figure 13: Parameter degeneracies. Left: radial position vs amplitude ($\Delta\chi^2 = 2.3, 6.2, 11.8$ contours, i.e. 1, 2, 3σ for two parameters): a farther shell needs more amplitude, giving an extended valley. Right: spectral slope vs amplitude at the chosen window. Degeneracies mean individual parameters are less well constrained than the combination.

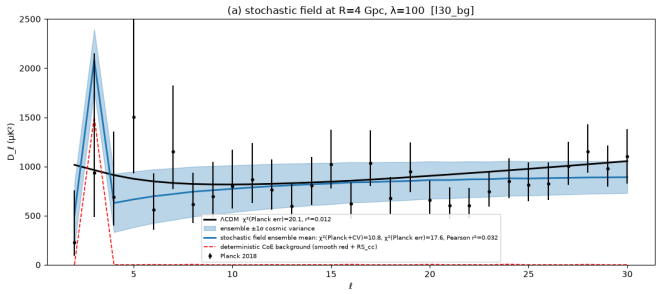


Figure 14: Acoustic-feature stochastic field, $R = 4\,\mathrm{Gpc}$, free_W: model peaks [227, 530, 831, 1131, 1431] vs Planck [224, 535, 821, 1122, 1412]; red band $\pm 1\sigma$ cosmic variance, black ΛCDM.

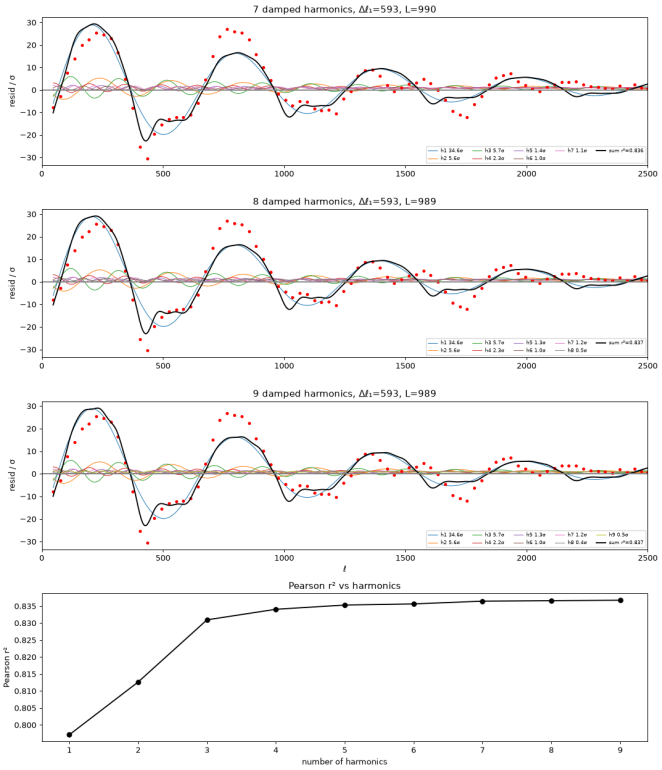


Figure 15: Damped-harmonic fits to ϵ_ℓ with 7, 8, 9 harmonics ($\Delta\ell = 593$, $L = 990$), and Pearson r^2 versus number of harmonics (bottom), rising from 0.797 to 0.837.

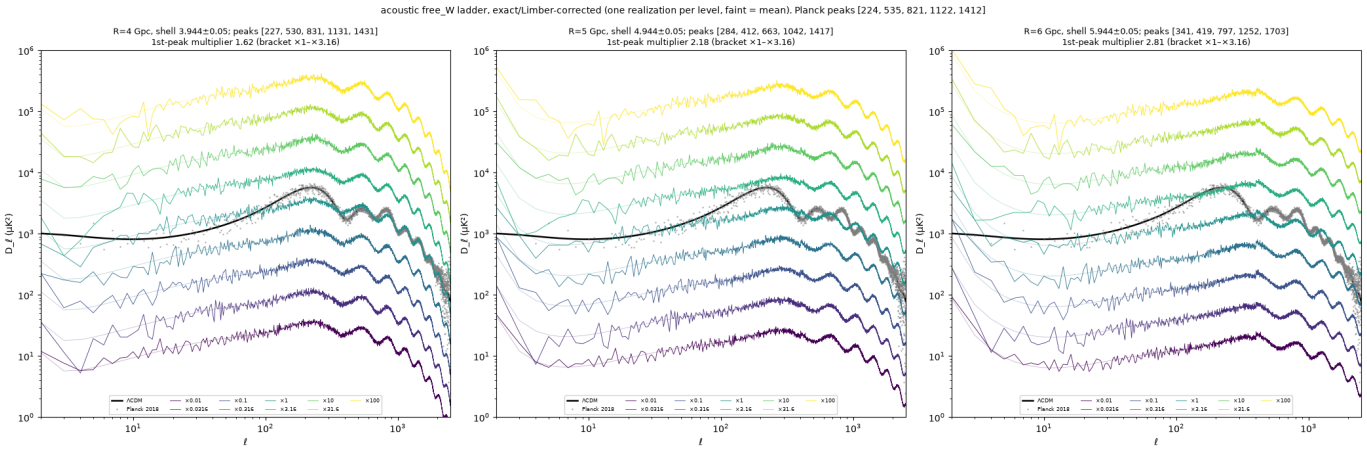


Figure 16: Acoustic free_W ladder (exact-limber corrected, one realisation per panel; Planck peaks 224, 535, 821, 1122, 1412). Panels: $R = 4$, 5, 6 Gpc with shell at 3.944, 4.944, 5.944 Gpc; model peaks [227, 530, 831, 1131, 1431], [284, 412, 663, 1042, 1417], [341, 419, 797, 1252, 1703] at fixed source parameters. The peak spacing is set by r_s and R_c jointly ($\Delta\ell \approx 2\pi R_c/r_s$), so moving the shell without refitting r_s shifts the peaks.

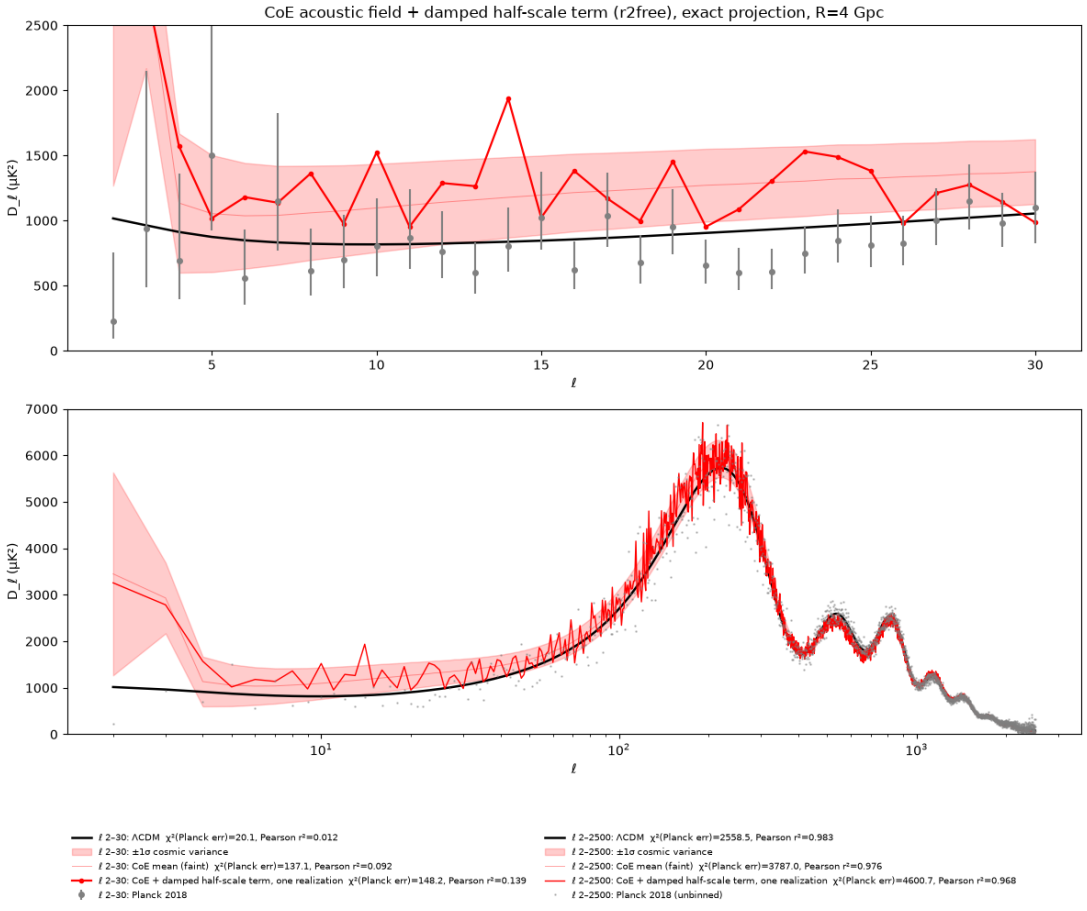


Figure 17: Final model: CoE acoustic field + damped half-scale term (r_2 free), exact projection, $R \approx 4$ Gpc. Top: low ℓ (linear). Bottom: full spectrum, ΛCDM black, Planck grey, CoE one-sky realisation red in its $\pm 1\sigma$ band with faint mean. $\chi^2_{2-2500} = 3{,}787$, Pearson $r^2 = 0.976$.